The Valuation of European Options in Uncertain Environment

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Abstract

A new model on European options with uncertainty of both randomness and fuzziness in output is presented, by introducing fuzzy logic to the stochastic financial model. The randomness and fuzziness in the systems are evaluated by both probabilistic expectation and fuzzy expectation, taking account of sellers/buyers subjective judgment. Prices of European call/put options with uncertainty are given and their valuation and properties are discussed under a rea sonable assumption. This paper demonstrates Black–Scholes formula to give rational expected price of the European options and buyers/writers (sellers) permissible range of expected prices. The meaning and properties of rational expected prices are discussed in a numerical example. The hedging strategies are also considered for marketability of the European options for portfolio selection.

1 Introduction

The theory of option pricing has developed primarily through the Black–Scholes log-normal stochastic model. However, when we trade stocks in practice, there is often a difference between the actual market price and the theoretical value calculated from Black–Scholes methods. When markets change rapidly, this gap between the expected price and actual price becomes larger.

In the classical framework, the stock price process $\{S_t\}_{t\geq 0}$ follows the stochastic differential equation

$$dS_t = \mu S_t\, dt + \sigma S_t\, dB_t,$$

where μ is the drift rate, σ is the volatility, and B_t is a standard Brownian motion. While this model gives us elegant formulas, real markets show rapid fluctuations and sudden changes that cause actual prices to deviate from theoretical predictions.

Classical stochastic models handle uncertainty using only probability theory, but in practice, uncertainty also arises from our inability to identify exact future values due to incomplete market knowledge, a type of ambiguity better described as *fuzziness* rather than randomness.

1.1 Fuzziness in Financial Markets

Fuzzy logic extends classical mathematics to handle imprecise information. Instead of representing a value by a single number, we use fuzzy numbers with membership functions. A fuzzy number $\tilde{x}$ is characterized by its membership function $\mu_{\tilde{x}} : \mathbb{R} \to [0, 1]$. For example, a triangular fuzzy number has the form

$$\tilde{x} = (x_1, x_2, x_3),$$

where x_1 and x_3 represent the bounds of uncertainty and x_2 represents the most likely value.

In financial markets, fuzziness reflects the difficulty in identifying exact values due to incomplete information about future volatility, uncertain investor beliefs, vague economic conditions, and data limitations. To handle this, we extend classical random variables to *fuzzy random variables*, which take fuzzy number values rather than single real numbers.

1.2 Fuzzy Stock Price Modeling

In our fuzzy stochastic model, the bond price remains deterministic and satisfies

$$dR_t = rR_t\, dt, \qquad R_0 = 1,$$

which gives the solution $R_t = e^{rt}$ for $t \geq 0$.

The stock price still follows the log-normal stochastic differential equation

$$dS_t = \mu S_t\, dt + \sigma S_t\, dB_t,$$

but now we interpret S_t as a *fuzzy number*. This means the stock price carries both randomness from the Brownian motion B_t and fuzziness arising from uncertain market conditions and imprecise information.

To evaluate fuzzy-valued processes, we use fuzzy expectation based on a possibility measure Π, which generalizes classical expectation. For a fuzzy random variable $\tilde{X}$, the fuzzy expectation at confidence level α is defined through α-cuts as

$$E[\tilde{X}]_\alpha = \left[E[X_\alpha^-],\, E[X_\alpha^+]\right], \qquad 0 \leq \alpha \leq 1,$$

where $[X_\alpha^-, X_\alpha^+]$ represents the α-level set of $\tilde{X}$.

1.3 European Options Under Fuzzy Uncertainty

In the fuzzy extension of the Black–Scholes model, the payoff of a European call option,

$$C = \max(S_T - K, 0),$$

becomes a fuzzy random variable when the terminal stock price S_T is fuzzy-valued. Instead of obtaining a single precise option price, we get a confidence interval of acceptable prices. This interval reflects both the randomness of market movements and the uncertainty in identifying exact parameter values.

This fuzzy pricing approach provides a realistic range of acceptable option prices, establishes permissible price intervals for both buyers and sellers, and naturally incorporates subjective judgment into valuation decisions.

1.4 Role of PDEs and Computational Methods

The classical Black–Scholes PDE for a European option price $V(S,t)$ is given by

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

When we introduce fuzziness into the model, parameters such as r, σ, and even S may become fuzzy numbers. The PDE must then be solved with fuzzy-valued inputs, and numerical methods like finite difference schemes need to be adapted to propagate fuzziness through the computation. This makes fuzzy option pricing an important and challenging computational problem in financial mathematics.

2 Fuzzy Stochastic Processes

Financial markets involve both random behavior and uncertainty in the information used to describe them. Classical probability models capture randomness but not the imprecision present in many market parameters. Introducing fuzzy random variables allows us to represent both effects within a single framework.

2.1 Fuzzy Numbers and Operations

A *fuzzy number* is characterized by its membership function $\tilde{a} : \mathbb{R} \to [0,1]$, which must satisfy normality (there exists $x_0 \in \mathbb{R}$ such that $\tilde{a}(x_0) = 1$), upper semicontinuity, fuzzy convexity (all α-cuts form convex sets), and compact support (the closure of $\{x \mid \tilde{a}(x) > 0\}$ is bounded). We denote the collection of all fuzzy numbers by $\mathcal{F}(\mathbb{R})$. The value $\tilde{a}(x)$ represents the degree of membership or confidence that the real number x belongs to the fuzzy set $\tilde{a}$.

The α-cut of a fuzzy number $\tilde{a}$ at level $\alpha \in (0,1]$ captures all values with membership at least α:

$$\tilde{a}_\alpha := \{x \in \mathbb{R} \mid \tilde{a}(x) \geq \alpha\}.$$

For $\alpha = 0$, we define $\tilde{a}_0 := \mathrm{cl}\{x \in \mathbb{R} \mid \tilde{a}(x) > 0\}$, where cl denotes closure. Since fuzzy numbers have the convexity property, each α-cut forms a closed interval:

$$\tilde{a}_\alpha = [\tilde{a}_\alpha^-, \tilde{a}_\alpha^+] \quad \text{for all } \alpha \in [0,1],$$

where $\tilde{a}_\alpha^-$ and $\tilde{a}_\alpha^+$ denote the left and right endpoints respectively.

Arithmetic operations on fuzzy numbers are defined through their α-cuts. For fuzzy numbers $\tilde{a}, \tilde{b} \in \mathcal{F}(\mathbb{R})$ and scalar $\lambda \geq 0$, we define addition, subtraction, and scalar multiplication as

follows:

$$\begin{aligned}(\tilde{a}+\tilde{b})_\alpha &= [\tilde{a}_\alpha^- + \tilde{b}_\alpha^-, \tilde{a}_\alpha^+ + \tilde{b}_\alpha^+],\\ (\tilde{a}-\tilde{b})_\alpha &= [\tilde{a}_\alpha^- - \tilde{b}_\alpha^+, \tilde{a}_\alpha^+ - \tilde{b}_\alpha^-],\\ (\lambda\tilde{a})_\alpha &= [\lambda\tilde{a}_\alpha^-, \lambda\tilde{a}_\alpha^+].\end{aligned}$$

We introduce the fuzzy max order $\preceq$ on $\mathcal{F}(\mathbb{R})$ as a partial ordering. For fuzzy numbers $\tilde{a}, \tilde{b}$, we write $\tilde{a} \preceq \tilde{b}$ if and only if

$$\tilde{a}_\alpha^- \geq \tilde{b}_\alpha^- \quad \text{and} \quad \tilde{a}_\alpha^+ \geq \tilde{b}_\alpha^+ \quad \text{for all } \alpha \in [0,1].$$

Under this ordering, the space $(\mathcal{F}(\mathbb{R}), \preceq)$ becomes a lattice. The fuzzy maximum $\tilde{a} \vee \tilde{b}$ with respect to this order is defined through its α-cuts:

$$(\tilde{a} \vee \tilde{b})_\alpha = [\max\{\tilde{a}_\alpha^-, \tilde{b}_\alpha^-\}, \max\{\tilde{a}_\alpha^+, \tilde{b}_\alpha^+\}].$$

2.2 Fuzzy Random Variables

Let $(\Omega, \mathcal{F}, P)$ be a probability space where $\mathcal{F}$ is a σ-field and P is a probability measure. A fuzzy random variable is a mapping $\tilde{X} : \Omega \to \mathcal{F}(\mathbb{R})$ satisfying the measurability condition

$$\{(\omega, x) \in \Omega \times \mathbb{R} \mid \tilde{X}(\omega)(x) \geq \alpha\} \in \mathcal{F} \otimes \mathcal{B} \quad \text{for all } \alpha \in [0,1],$$

where $\mathcal{B}$ denotes the Borel σ-field of $\mathbb{R}$. This definition ensures that fuzzy random variables are compatible with the underlying probability structure and can be properly integrated.

The following lemma provides a practical characterization that simplifies verification and computation in applications.

Lemma 2.1. *A mapping $\tilde{X} : \Omega \to \mathcal{F}(\mathbb{R})$ is a fuzzy random variable if and only if the endpoint functions*

$$\omega \mapsto \tilde{X}_\alpha^-(\omega) \quad \textit{and} \quad \omega \mapsto \tilde{X}_\alpha^+(\omega)$$

are measurable for all $\alpha \in [0,1]$, where $\tilde{X}_\alpha(\omega) = [\tilde{X}_\alpha^-(\omega), \tilde{X}_\alpha^+(\omega)]$.

This characterization is particularly useful because it reduces the verification of measurability to checking the endpoints of interval-valued functions rather than working directly with the membership function itself.

2.3 Expectations of Fuzzy Random Variables

For an integrably bounded fuzzy random variable $\tilde{X}$ (where the endpoint functions $\omega \mapsto \tilde{X}_\alpha^\pm(\omega)$ are integrable for all $\alpha \in [0,1]$), the expectation $E(\tilde{X})$ is defined as a fuzzy number whose α-cuts are given by

$$E(\tilde{X})_\alpha = \left[\int_\Omega \tilde{X}_\alpha^-(\omega)\, dP(\omega), \int_\Omega \tilde{X}_\alpha^+(\omega)\, dP(\omega)\right].$$

This definition naturally extends the classical probabilistic expectation to the fuzzy setting while preserving the linearity property. The membership function of the expected value $E(\tilde{X})$ can be reconstructed from these α-cuts using the representation

$$E(\tilde{X})(x) = \sup_{\alpha \in [0,1]} \min\{\alpha, \mathbb{1}_{E(\tilde{X})_\alpha}(x)\},$$

where $\mathbb{1}_A$ denotes the indicator function of set A.

e underlying probability space.

3 European Options in Uncertain Environment

We now apply the fuzzy stochastic framework to European option pricing, providing a rigorous extension of the Black-Scholes model that incorporates parameter uncertainty while maintaining analytical tractability.

3.1 Classical Stock Price Model

Under the risk-neutral measure Q, the classical stock price process $\{S_t\}_{t\geq 0}$ follows geometric Brownian motion governed by the stochastic differential equation

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t,$$

where $r \geq 0$ is the risk-free interest rate, $\sigma > 0$ is the volatility parameter, and $\{W_t\}_{t\geq 0}$ is a standard Brownian motion under Q. Application of Itô's formula yields the explicit solution

$$S_t = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)t + \sigma W_t\right),$$

which shows that S_t follows a log-normal distribution under the risk-neutral measure.

3.2 Fuzzy Stock Price Process

To model uncertainty arising from imprecise knowledge of the volatility parameter, we introduce a fuzzy stock price process. Let $\{a_t\}_{t\geq 0}$ be an $\mathcal{F}_t$-adapted stochastic process satisfying $0 < a_t(\omega) \leq S_t(\omega)$ for almost all $\omega \in \Omega$ and such that the mapping $t \mapsto a_t(\omega)$ is continuous on $[0, \infty)$. We define the fuzzy stock price process $\{\tilde{S}_t\}_{t\geq 0}$ through its membership function

$$\tilde{S}_t(\omega)(x) := L\left(\frac{x - S_t(\omega)}{a_t(\omega)}\right), \quad t \geq 0,$$

where $L(x) := \max\{1 - |x|, 0\}$ is the triangular shape function. This construction centers the fuzzy stock price at the classical stock price $S_t(\omega)$ with a spread determined by $a_t(\omega)$.

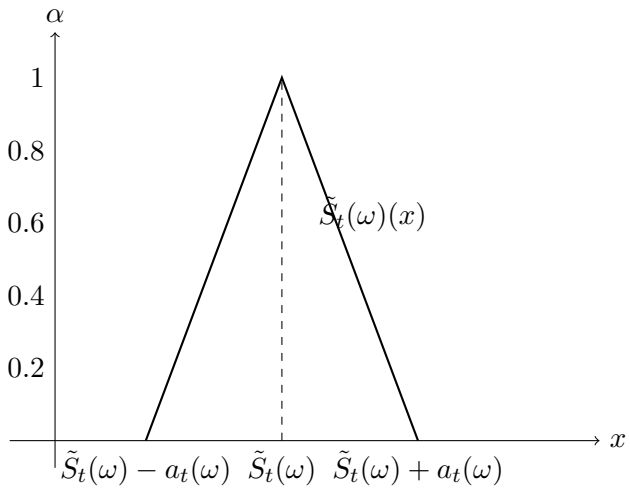

Figure 1: Fuzzy random variable $\tilde{S}_t(\omega)(x)$.

The α-cuts of the fuzzy stock price are given by

$$\tilde{S}_{t,\alpha}(\omega) = [S_t(\omega) - (1-\alpha)a_t(\omega), S_t(\omega) + (1-\alpha)a_t(\omega)].$$

The spread $a_t(\omega)$ represents the degree of uncertainty about the stock price, with higher values of α corresponding to higher confidence levels and consequently yielding narrower intervals around the classical price $S_t(\omega)$.

We now introduce a fundamental assumption that ensures tractability while maintaining economic realism.

Assumption 1 (Proportional Spread)**.** *The uncertainty spread is proportional to the stock price:*

$$a_t(\omega) = cS_t(\omega), \quad t \geq 0,$$

where $c \in (0,1)$ is a constant called the fuzzy factor.

This assumption is economically justified by observing that uncertainty should scale with price levels, analogous to how volatility enters multiplicatively in the stochastic differential equation for S_t. Since one of the primary difficulties in practical applications is the estimation of the volatility parameter σ, it is natural that the spread $a_t(\omega)$ should depend on both the volatility and the stock price through the term $\sigma S_t(\omega)$ appearing in the diffusion coefficient. The fuzzy factor c quantifies the degree of volatility uncertainty in the model.

Under Assumption 1, we define the scaling factors $b^{\pm}(\alpha) := 1 \mp (1-\alpha)c$ for $\alpha \in [0,1]$. This yields the simplified representation

$$\tilde{S}^{\pm}_{t,\alpha}(\omega) = b^{\pm}(\alpha)S_t(\omega),$$

which shows that at each confidence level α, the fuzzy stock price endpoints are deterministic multiples of the classical stock price. As $\alpha \to 1$, we have $b^{\pm}(\alpha) \to 1$, recovering the classical crisp stock price, while as $\alpha \to 0$, the interval achieves its maximum width, reflecting maximum uncertainty.

3.3 Fuzzy Option Payoffs

For a European call option with strike price $K > 0$ and maturity $T > 0$, we define the fuzzy payoff at time t by

$$\tilde{C}_t(\omega) := e^{-rt}(\tilde{S}_t(\omega) - K) \vee 0,$$

where the fuzzy maximum $\vee$ is defined with respect to the fuzzy max order introduced in Section 2.1 .

The α-cuts of this fuzzy payoff are

$$\tilde{C}_{t,\alpha}(\omega) = [\max\{e^{-rt}(\tilde{S}^{-}_{t,\alpha}(\omega) - K), 0\}, \max\{e^{-rt}(\tilde{S}^{+}_{t,\alpha}(\omega) - K), 0\}].$$

Similarly, for a European put option, we define

$$\tilde{P}_t(\omega) := e^{-rt}(K - \tilde{S}_t(\omega)) \vee 0,$$

with corresponding α-cuts

$$\tilde{P}_{t,\alpha}(\omega) = [\max\{e^{-rt}(K - \tilde{S}^{+}_{t,\alpha}(\omega)), 0\}, \max\{e^{-rt}(K - \tilde{S}^{-}_{t,\alpha}(\omega)), 0\}].$$

3.4 Fuzzy Option Prices

The fuzzy price processes at time t with current stock price $y = S_t$ are defined as conditional expectations under the risk-neutral measure Q:

$$\tilde{V}_{\tilde{C}}(y,t) := e^{-rt}E^Q[\tilde{C}_T \mid S_t = y] = E^Q[e^{-r(T-t)}(\tilde{S}_T - K) \vee 0 \mid S_t = y],$$
$$\tilde{V}_{\tilde{P}}(y,t) := e^{-rt}E^Q[\tilde{P}_T \mid S_t = y] = E^Q[e^{-r(T-t)}(K - \tilde{S}_T) \vee 0 \mid S_t = y].$$

These definitions extend the classical risk-neutral valuation principle to the fuzzy setting by taking expectations of fuzzy payoffs.

The α-cut endpoints of the fuzzy call and put prices at time $t = 0$ are given by

$$\tilde{V}^{\pm}_{\tilde{C},\alpha}(y,0) = E^Q[e^{-rT}\max\{b^{\pm}(\alpha)S_T - K, 0\} \mid S_0 = y],$$
$$\tilde{V}^{\pm}_{\tilde{P},\alpha}(y,0) = E^Q[e^{-rT}\max\{K - b^{\pm}(\alpha)S_T, 0\} \mid S_0 = y].$$

These expressions show that pricing fuzzy options reduces to pricing classical options with modified initial stock prices $b^{\pm}(\alpha)y$.

3.5 Black-Scholes Formulas for Fuzzy Prices

We now state the main pricing results for fuzzy European options.

Theorem 3.1 (Fuzzy Black-Scholes Formula)**.** *Under Assumption 1, the fuzzy call option price at $t = 0$ with initial stock price $y > 0$ has α-cut endpoints given by*

$$\tilde{V}^{\pm}_{\tilde{C},\alpha}(y,0) = b^{\pm}(\alpha)y\Phi(z_1^{\pm}) - Ke^{-rT}\Phi(z_2^{\pm}),$$

where Φ is the standard normal cumulative distribution function and

$$z_1^{\pm} = \frac{\log b^{\pm}(\alpha) + \log(y/K) + T(r + \sigma^2/2)}{\sigma\sqrt{T}},$$
$$z_2^{\pm} = z_1^{\pm} - \sigma\sqrt{T}.$$

The proof follows the classical derivation by noting that under the risk-neutral measure, $b^{\pm}(\alpha)S_T$ has the same log-normal distribution as S_T scaled by the constant factor $b^{\pm}(\alpha)$.

Theorem 3.2 (Call-Put Parity)**.** *The fuzzy put option price satisfies the parity relation*

$$\tilde{V}^{\pm}_{\tilde{P},\alpha}(y,0) = \tilde{V}^{\pm}_{\tilde{C},\alpha}(y,0) - b^{\pm}(\alpha)y + Ke^{-rT}.$$

Proof. Since $(e^{-rt}S_t)_{t\geq 0}$ is a martingale under Q, we have

$$\begin{aligned}
\tilde{V}^{\pm}_{\tilde{P},\alpha}(y,0)/b^{\pm}(\alpha) &= E^Q\left[e^{-rT}\max\{K/b^{\pm}(\alpha) - S_T, 0\} \mid S_0 = y\right] \\
&= E^Q\left[e^{-rT}\max\{S_T - K/b^{\pm}(\alpha), 0\} \mid S_0 = y\right] \\
&\quad - E^Q[e^{-rT}S_T \mid S_0 = y] + Ke^{-rT}/b^{\pm}(\alpha) \\
&= \tilde{V}^{\pm}_{\tilde{C},\alpha}(y,0)/b^{\pm}(\alpha) - E^Q[S_0 \mid S_0 = y] + Ke^{-rT}/b^{\pm}(\alpha) \\
&= \tilde{V}^{\pm}_{\tilde{C},\alpha}(y,0)/b^{\pm}(\alpha) - y + Ke^{-rT}/b^{\pm}(\alpha).
\end{aligned}$$

Multiplying both sides by $b^{\pm}(\alpha)$ yields the desired result. □

4 Expected Prices of European Options

The expected price framework builds on fuzzy stochastic processes where stock prices take fuzzy values rather than crisp numbers. The key innovation lies in how these prices are evaluated using a dual assessment mechanism: probabilistic expectation handles randomness while fuzzy expectation addresses vagueness in price identification.

The expected price analysis begins with a fixed initial stock price $y > 0$ and considers European call/put options with fuzzy prices denoted by $\tilde{V} = \tilde{V}_{\tilde{C}}(y,0)$ or $\tilde{V} = \tilde{V}_{\tilde{P}}(y,0)$.

4.1 Grade of Fuzzy Expectation for Call Options

For European call options, we define a critical grade $\alpha_{\tilde{C},+}$ that characterizes the fuzzy expectation:

Call Option Grade: The grade $\alpha_{\tilde{C},+}$ is defined as:

$$\alpha_{\tilde{C},+} := \sup\{\alpha \in [0,1] \mid u_\alpha^- \leq \tilde{V}_{\tilde{C},+}^{\alpha}(y,0)\}$$

where $u_\alpha = [u_\alpha^-, \infty)$ for $\alpha \in (0,1)$ represents the α-cut of the fuzzy goal u.

Due to the continuity properties of u and $\tilde{V}_{\tilde{C}}$, the grade satisfies:

$$u_{\alpha_{\tilde{C},+}}^- = \tilde{V}_{\tilde{C},+}^{\alpha_{\tilde{C},+}}(y,0)$$

Theorem 4.1 (Fuzzy Expectation of Call Options)**.** *Under the fuzzy expectation generated by possibility measures, the following hold:*

(i) The grade of the fuzzy expectation of European call option price is:

$$\alpha_{\tilde{C},+} = \tilde{E}(\tilde{V}_{\tilde{C}}(y,0)) = \tilde{E}(E(\tilde{C}_T) \mid S_0 = y)$$

(ii) The rational expected price of European call option is:

$$x_{\tilde{C},+} = u_{\alpha_{\tilde{C},+}}^-$$

This rational expected price represents an upper bound on expected prices from the buyer's perspective, as the fuzzy expectation is defined using possibility measures.

4.2 Permissible Price Range

To establish a complete confidence interval, we define a lower bound for rational expected prices:

$$x_{\tilde{C},-} = u_{\alpha_{\tilde{C},-}}^-$$

where the grade $\alpha_{\tilde{C},-}$ is defined by:

$$\alpha_{\tilde{C},-} := \sup\{\alpha \in [0,1] \mid u_\alpha^- \leq \tilde{V}_{\tilde{C},-}^{\alpha}(y,0)\}$$

satisfying:

$$u_{\alpha_{\tilde{C},-}}^- = \tilde{V}_{\tilde{C},-}^{\alpha_{\tilde{C},-}}(y,0)$$

Buyer's Permissible Range: The buyer's permissible range of expected prices is given by:

$$[x_{\tilde{C},-}, x_{\tilde{C},+}] = \{x \in \mathbb{R} \mid \tilde{V}_{\tilde{C}}(y,0)(x) \geq u(x)\}$$

This interval represents the range of prices where the reliability degree of the optimal expected price exceeds the degree of buyer's satisfaction under their fuzzy goal u.

This formulation has practical significance: it provides buyers and sellers with confidence intervals for negotiation, acknowledging that uncertainty cannot be reduced to a point estimate.

4.3 Put Options and Writer's Perspective

Similarly, for European put options, the writer's permissible range of rational expected prices is:

$$[x_{\tilde{P},-}, x_{\tilde{P},+}]$$

where:

$$x_{\tilde{P},-} := u^{-}_{\alpha_{\tilde{P},-}}$$
$$x_{\tilde{P},+} := u^{-}_{\alpha_{\tilde{P},+}}$$

with grades determined by:

$$u^{-}_{\alpha_{\tilde{P},-}} = \tilde{V}^{\alpha_{\tilde{P},-}}_{\tilde{P},-}(y, 0)$$
$$u^{-}_{\alpha_{\tilde{P},+}} = \tilde{V}^{\alpha_{\tilde{P},+}}_{\tilde{P},+}(y, 0)$$

4.4 Practical Implications

The permissible price ranges provide several practical benefits:

1. **Risk Management**: Buyers and sellers can assess price reliability under their subjective goals
2. **Decision Support**: The intervals guide negotiation and pricing strategies
3. **Uncertainty Quantification**: The ranges explicitly account for both randomness and fuzziness
4. **Flexibility**: Different fuzzy goals u can represent varying risk preferences

Remark. The methodology assumes Assumption S (fuzzy factor proportional to stock price) and triangular membership functions for computational tractability. While these assumptions can be relaxed, they provide reasonable approximations for practical applications.

5 Hedging Strategies

Hedging strategies are essential for managing risk in option portfolios. This section develops optimal hedging strategies for European call options under fuzzy uncertainty.

5.1 Basic Definitions

Fix any $\alpha \in [0, 1]$. A hedging strategy is defined as an $\mathcal{M}_t$-predictable process:

Hedging Strategy: A hedging strategy is a process $\{(\pi^t_{0,\pm}, \pi^t_{1,\pm})\}_{t\geq 0}$ with values in $\mathbb{R} \times \mathbb{R}$, where:

- $\pi^t_{0,\pm}$ represents the amount invested in the bond at time t
- $\pi^t_{1,\pm}$ represents the amount invested in the stock at time t

The strategy satisfies:

$$V^{t,\alpha}_{\pm} = \pi^t_{0,\pm} R_t + \pi^t_{1,\pm} \tilde{S}^{t,\alpha}_{\pm}, \quad t \geq 0$$

where $V^{t,\alpha}_{\pm} := e^{rt} E(\tilde{C}^{T,\alpha}_{\pm} \mid \mathcal{M}_t)$ is the wealth process.

5.2 Self-Financing Condition

Self-Financing Strategy: A hedging strategy $\{(\pi^t_{0,\pm}, \pi^t_{1,\pm})\}_{t\geq 0}$ is called self-financing if:

$$d\pi^t_{0,\pm} R_t + d\pi^t_{1,\pm} \tilde{S}^{t,\alpha}_{\pm} = 0, \quad t \geq 0$$

and

$$\int_0^T |\pi^t_{0,\pm}| dt < \infty \quad \text{and} \quad \int_0^T (\pi^t_{1,\pm})^2 dt < \infty \quad \text{almost surely}$$

The self-financing condition ensures no external capital inflow or outflow during the hedging period. This property is crucial for practical portfolio management.

For a self-financing strategy, the wealth dynamics are:

$$dV^{t,\alpha}_{\pm} = \pi^t_{0,\pm} dR_t + \pi^t_{1,\pm} d\tilde{S}^{t,\alpha}_{\pm}, \quad t \geq 0$$

Integrating this expression yields:

$$V^{t,\alpha}_{\pm} = \pi^0_{0,\pm} R_0 + \pi^0_{1,\pm} \tilde{S}^{0,\alpha}_{\pm} + \int_0^t \pi^u_{0,\pm} dR_u + \int_0^t \pi^u_{1,\pm} d\tilde{S}^{u,\alpha}_{\pm}, \quad t \geq 0$$

5.3 Minimal Hedging Strategy

Theorem 5.1 (Optimal Hedging Strategy)**.** *The minimal hedging strategy* $\{(\pi^t_{0,\pm}, \pi^t_{1,\pm})\}_{t\in[0,T]}$ *for the fuzzy price of European call option is given by:*

Bond position:

$$\pi^t_{0,\pm} = \Phi\left(\frac{\log b_{\pm}(\alpha) + \log(S_t/K) + (T-t)(r + \sigma^2/2)}{\sigma\sqrt{T-t}}\right)$$

Stock position:

$$\pi^t_{1,\pm} = -e^{-rT} K \Phi\left(\frac{\log b_{\pm}(\alpha) + \log(S_t/K) + (T-t)(r - \sigma^2/2)}{\sigma\sqrt{T-t}}\right)$$

for $t < T$, where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function.

The corresponding wealth process is:

$$V^{t,\alpha}_{\pm} = \pi^t_{0,\pm} R_t + \pi^t_{1,\pm} \tilde{S}^{t,\alpha}_{\pm}$$

5.4 Properties of the Hedging Strategy

5.4.1 Delta Hedging Interpretation

The stock position $\pi^t_{1,\pm}$ represents the option's delta, measuring sensitivity to stock price changes. Key properties include:

1. **Dynamic Adjustment**: The positions must be continuously rebalanced as t and S_t change
2. **Boundary Behavior**: As $t \to T$, the strategy converges to discrete payoff replication
3. **Monotonicity**: The delta increases with the stock price for call options

5.4.2 Risk Management Implications

The minimal hedging strategy provides:

- **Complete Replication**: The wealth process exactly matches the option payoff at expiration
- **Cost Minimization**: Among all replicating strategies, this requires minimum initial capital
- **Market Neutrality**: The self-financing property eliminates directional risk

5.5 Comparison with Classical Approach

Feature	**Classular Hedging**	**Fuzzy Hedging**
Uncertainty type	Randomness only	Randomness + Fuzziness
Price output	Single value	Fuzzy interval
Strategy uniqueness	Unique delta	α-parameterized family
Computational complexity	Moderate	Higher
Market applicability	Stable markets	Volatile/uncertain markets

Table 1: Comparison of hedging approaches

The fuzzy approach provides additional flexibility for decision makers facing epistemic uncertainty beyond stochastic volatility.

References

[1] Y. Yoshida, *The valuation of European options in uncertain environment*, European Journal of Operational Research 145 (2003) 221–229.